

## ● RAG Learning

# 1.2 Algebra

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs a1–a15

**What this topic covers.** The algebraic toolkit the rest of the A-Level runs on: surds and indices, quadratics and the discriminant, simultaneous equations, inequalities and set notation, proportion, and partial fractions.

## 1. Algebraic language

MEI expects the vocabulary to be used precisely (spec ref a1).

| Term                      | Meaning                                                                                |
|---------------------------|----------------------------------------------------------------------------------------|
| <b>Expression</b>         | A collection of terms, with no equals sign: $3x^2 - 5x$ .                              |
| <b>Equation</b>           | Two expressions set equal, true only for particular values: $x^2 = 9$ .                |
| <b>Identity</b>           | True for <i>every</i> value; written with $\equiv$ : $(x + 1)^2 \equiv x^2 + 2x + 1$ . |
| <b>Function</b>           | A rule assigning exactly one output to each input.                                     |
| <b>Coefficient</b>        | The numerical multiplier of a term: in $4x^3$ , it is 4.                               |
| <b>Term</b>               | A single part of an expression, separated by $+$ or $-$ .                              |
| <b>Variable / unknown</b> | A quantity that varies / a fixed value to be found.                                    |

## 2. Surds

A **surd** is an irrational root left in exact form. Working in surds keeps answers exact; decimals introduce rounding error.

## The rules

$$\sqrt{ab} = \sqrt{a} \sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad (\sqrt{a})^2 = a$$

Simplify by extracting the largest square factor:

$$\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}, \quad \sqrt{8} + \sqrt{2} = 2\sqrt{2} + \sqrt{2} = 3\sqrt{2}$$

$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ . Test it:  $\sqrt{9+16} = 5$ , but  $\sqrt{9} + \sqrt{16} = 7$ . The product rule holds; the sum rule does not.

## Rationalising the denominator

For a single surd, multiply top and bottom by that surd:

$$\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

For a denominator  $a + \sqrt{b}$ , multiply by its **conjugate**  $a - \sqrt{b}$ , using the difference of two squares to clear the surd:

**Worked example.** Rationalise  $\frac{1}{3 + \sqrt{5}}$ .

$$\frac{1}{3 + \sqrt{5}} \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}} = \frac{3 - \sqrt{5}}{9 - 5} = \frac{3 - \sqrt{5}}{4}$$

**Worked example.** Rationalise  $\frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} + \sqrt{3}}$ .

$$\frac{(\sqrt{7} - \sqrt{3})^2}{(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})} = \frac{7 - 2\sqrt{21} + 3}{7 - 3} = \frac{10 - 2\sqrt{21}}{4} = \frac{5 - \sqrt{21}}{2}$$

## 3. Indices

| Law                         | Example                |
|-----------------------------|------------------------|
| $x^a \times x^b = x^{a+b}$  | $x^3 x^4 = x^7$        |
| $x^a \div x^b = x^{a-b}$    | $x^5 \div x^2 = x^3$   |
| $(x^a)^b = x^{ab}$          | $(x^2)^3 = x^6$        |
| $x^0 = 1$ (for $x \neq 0$ ) | $7^0 = 1$              |
| $x^{-a} = \frac{1}{x^a}$    | $2^{-3} = \frac{1}{8}$ |
| $x^{1/n} = \sqrt[n]{x}$     | $8^{1/3} = 2$          |
| $x^{m/n} = (\sqrt[n]{x})^m$ | $8^{2/3} = 2^2 = 4$    |

For  $x^{m/n}$ , take the **root first**, then the power — the numbers stay small.  $8^{2/3}$  as  $(\sqrt[3]{8})^2 = 4$  is far easier than  $\sqrt[3]{64}$ .

## 4. Quadratic equations

Three routes, plus the graphical picture (spec refs a2, a6).

### Factorising

$$x^2 - 5x + 6 = 0 \implies (x - 2)(x - 3) = 0 \implies x = 2 \text{ or } x = 3$$

### Completing the square

$$x^2 + 6x + 2 = (x + 3)^2 - 9 + 2 = (x + 3)^2 - 7$$

This form reads off the vertex directly: the minimum of  $y = (x + 3)^2 - 7$  is at  $(-3, -7)$ .

### The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## Quadratics in disguise

**Worked example.** Solve  $2^{2x} - 5 \cdot 2^x + 4 = 0$ .

Let  $u = 2^x$ . Since  $2^{2x} = (2^x)^2 = u^2$ :

$$u^2 - 5u + 4 = 0 \implies (u - 1)(u - 4) = 0 \implies u = 1 \text{ or } u = 4$$

So  $2^x = 1$  gives  $x = 0$ , and  $2^x = 4$  gives  $x = 2$ .

**Worked example.** Solve  $x^4 - 5x^2 + 4 = 0$ .

Let  $u = x^2$ :  $u^2 - 5u + 4 = 0$ , so  $u = 1$  or  $u = 4$ , giving  $x = \pm 1$  or  $x = \pm 2$  — four roots.

## 5. The discriminant

For  $ax^2 + bx + c = 0$ , the **discriminant** is  $\Delta = b^2 - 4ac$  (spec ref a3).

| Discriminant    | Roots                   | Graph and the $x$ -axis |
|-----------------|-------------------------|-------------------------|
| $b^2 - 4ac > 0$ | Two distinct real roots | Crosses twice           |
| $b^2 - 4ac = 0$ | One repeated root       | Touches (tangent)       |
| $b^2 - 4ac < 0$ | No real roots           | Never meets it          |

**Worked example.** Find the values of  $k$  for which  $kx^2 + 4x + k = 0$  has a repeated root.

$$b^2 - 4ac = 16 - 4k^2 = 0 \implies k^2 = 4 \implies k = \pm 2$$

**Worked example.** Find the values of  $k$  for which  $kx^2 + (k + 3)x + 4 = 0$  has no real roots.

$$(k + 3)^2 - 16k < 0$$

$$k^2 + 6k + 9 - 16k < 0 \implies k^2 - 10k + 9 < 0 \implies (k - 1)(k - 9) < 0$$

A negative product means  $k$  lies between the roots:  $1 < k < 9$ .

"Two distinct roots" means  $> 0$ ; "two real roots" or "real roots" usually allows the repeated case, so  $\geq 0$ . Read the wording — this distinction decides whether your final answer uses  $<$  or  $\leq$ .

## Tangency

A line meets a curve where their equations are equal. Substituting and demanding  $\Delta = 0$  is the standard condition for the line to be a **tangent**.

**Worked example.** Find the condition for  $y = mx + c$  to be tangent to  $y = x^2$ .

$$x^2 = mx + c \implies x^2 - mx - c = 0$$

Tangency requires  $\Delta = 0$ :  $(-m)^2 - 4(1)(-c) = m^2 + 4c = 0$ .

## 6. Simultaneous equations

Two linear equations: eliminate or substitute (a4). One linear and one quadratic: **always substitute the linear into the quadratic** (a5).

**Worked example.** Solve  $y = x + 1$  and  $x^2 + y^2 = 25$ .

$$x^2 + (x + 1)^2 = 25 \implies 2x^2 + 2x + 1 = 25 \implies x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0 \implies x = -4 \text{ or } x = 3$$

Substituting back into  $y = x + 1$ : the solutions are  $(-4, -3)$  and  $(3, 4)$ .

Substitute back into the **linear** equation, not the quadratic — it is quicker and cannot introduce spurious roots. And always pair the values: writing " $x = 3$  or  $x = -4$ ,  $y = 4$  or  $y = -3$ " without pairing loses marks.

Geometrically the solutions are the **points of intersection** of the two graphs (a6). If the resulting quadratic has no real roots, the graphs do not meet.

## 7. Inequalities

### Linear

Solve as an equation, but **reverse the inequality sign when multiplying or dividing by a negative**.

### Quadratic

Factorise, find the critical values, then read the sign from a sketch (a8).

**Worked example.** Solve  $x^2 - 4x - 5 > 0$ .

$$(x - 5)(x + 1) > 0$$

Critical values  $x = -1$  and  $x = 5$ . The parabola opens upwards, so it is *above* the axis outside the roots:

$$x < -1 \quad \text{or} \quad x > 5$$

A quick sketch settles the direction every time. For an upward parabola:  $< 0$  between the roots,  $> 0$  outside them. Guessing gets it backwards half the time.

### Inequalities with an algebraic fraction

Do **not** multiply both sides by  $x - 3$ : its sign is unknown, so the inequality may or may not reverse. Instead collect on one side over a common denominator, or multiply by the *square*  $(x - 3)^2$ , which is always positive.

## Set notation

Solutions may be expressed with "and"/"or" or in set notation (a9):

- $x \leq 1$  or  $x \geq 4 \rightarrow \{x : x \leq 1\} \cup \{x : x \geq 4\}$
- $2 < x < 5 \rightarrow \{x : x > 2\} \cap \{x : x < 5\}$

Two separate intervals need **or** (union). Writing " $x < -1$  and  $x > 5$ " describes numbers that are simultaneously less than  $-1$  and greater than  $5$  — nothing at all.

## 8. Proportion

| Statement                           | Equation            | Graph                            |
|-------------------------------------|---------------------|----------------------------------|
| $y$ directly proportional to $x$    | $y = kx$            | Straight line through the origin |
| $y$ inversely proportional to $x$   | $y = \frac{k}{x}$   | Hyperbola                        |
| $y$ proportional to $x^n$           | $y = kx^n$          | Depends on $n$                   |
| $y$ inversely proportional to $x^2$ | $y = \frac{k}{x^2}$ | Inverse-square curve             |

**Worked example.**  $y$  is inversely proportional to  $x^2$ , and  $y = 4$  when  $x = 3$ . Find  $y$  when  $x = 6$ .

$$y = \frac{k}{x^2} \implies 4 = \frac{k}{9} \implies k = 36$$

$$x = 6 : \quad y = \frac{36}{36} = 1$$

## 9. Partial fractions

**Partial fractions** split a single algebraic fraction into a sum of simpler ones (spec ref a15). MEI restricts this to denominators with up to **three linear factors**, and includes **repeated (squared)**

**linear factors.**

| Denominator             | Form to assume                                            |
|-------------------------|-----------------------------------------------------------|
| $(x + a)(x + b)$        | $\frac{A}{x + a} + \frac{B}{x + b}$                       |
| $(x + a)(x + b)(x + c)$ | $\frac{A}{x + a} + \frac{B}{x + b} + \frac{C}{x + c}$     |
| $(x + a)^2(x + b)$      | $\frac{A}{x + a} + \frac{B}{(x + a)^2} + \frac{C}{x + b}$ |

**Worked example.** Express  $\frac{5x + 1}{(x + 1)(x - 2)}$  in partial fractions.

$$\frac{5x + 1}{(x + 1)(x - 2)} = \frac{A}{x + 1} + \frac{B}{x - 2}$$

Multiply through by  $(x + 1)(x - 2)$ :

$$5x + 1 = A(x - 2) + B(x + 1)$$

Substitute strategic values. Let  $x = 2$ :  $11 = 3B$ , so  $B = \frac{11}{3}$ . Let  $x = -1$ :  $-4 = -3A$ , so  $A = \frac{4}{3}$ .

$$\frac{5x + 1}{(x + 1)(x - 2)} = \frac{4}{3(x + 1)} + \frac{11}{3(x - 2)}$$

For a **repeated** factor  $(x + a)^2$  you need *both*  $\frac{A}{x + a}$  and  $\frac{B}{(x + a)^2}$ . Omitting the first term is the single most common error here — the system then has no solution and the algebra collapses.

Substituting the values that make each factor zero is fastest. If a term remains (as with a repeated factor), compare coefficients of the highest power, or substitute one more convenient value such as  $x = 0$ .

## 10. Common pitfalls

- Writing  $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ .
- Forgetting the negative root:  $x^2 = 16$  gives  $x = \pm 4$ , not just 4.
- Multiplying an inequality by a negative — or by an expression of unknown sign — without reversing or justifying.
- Joining two separate solution intervals with "and" instead of "or".
- Reading a quadratic inequality off the wrong side of the parabola.
- Omitting the  $\frac{A}{x+a}$  term for a repeated linear factor in partial fractions.
- Confusing "two distinct real roots" ( $\Delta > 0$ ) with "real roots" ( $\Delta \geq 0$ ).
- Losing the pairing of  $x$  and  $y$  values in simultaneous equations.